

OSSP – Practical 1

Created a file named Practical_1.c using nano, compiled it with gcc, and ran the program. On entering the command **ls**, the child process executes it and the parent waits before printing its PID.

```
manu@Manasvi:~/os_lab$ nano Practical_1.c
manu@Manasvi:~/os_lab$ gcc Practical_1.c
manu@Manasvi:~/os_lab$ ./a.out
Enter command: ls
Child PID = 821
Practical_1.c  a.out
Parent PID = 772
```

Ran the program again with the command **pwd** — the child prints the current directory and the parent resumes only after the child finishes.

```
manu@Manasvi:~/os_lab$ gcc Practical_1.c
manu@Manasvi:~/os_lab$ ./a.out
Enter command: pwd
Child PID = 873
/home/manu/os_lab
Parent PID = 872
manu@Manasvi:~/os_lab$ |
```

Used the **uname** command, which returns Linux — the operating system in use.

```
manu@Manasvi:~/os_lab$ uname
Linux
manu@Manasvi:~/os_lab$ |
```

Used the **lscpu** command to view CPU details — architecture, number of cores, threads, and processor model.

```
manu@Manasvi:~/os_lab$ uname
Linux
manu@Manasvi:~/os_lab$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
Model name:                  12th Gen Intel(R) Core(TM) i5-1235U
CPU family:                  6
Model:                       154
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                   1
Stepping:                    4
BogoMIPS:                    4991.99
Flags:                       fpu vme de pse tsc msr pae mce cx8
                             apic sep mtrr pge mca cmov pat pse3
                             6 clflush mmx fxsr sse sse2 ss ht s
                             yscall nx pdpe1gb rdtscp lm constan
                             t_tsc rep_good nopl xtopology tsc_r
                             eliable nonstop_tsc cpuid tsc_known
                             _freq pni pclmulqdq vmx ssse3 fma c
                             x16 pcid sse4_1 sse4_2 x2apic movbe
                             popcnt tsc_deadline_timer aes xsav
                             e avx f16c rdrand hypervisor lahf_l
                             m abm 3dnowprefetch ssbd ibrs ibpb
                             stibp ibrs_enhanced tpr_shadow ept
                             vpid ept_ad fsgsbase tsc_adjust bmi
                             1 avx2 smep bmi2 erms invpcid rdsee
                             d adx smap clflushopt clwb sha_ni x
                             saveopt xsavec xgetbv1 xsaves avx_v
                             nni vnmi umip waitpkg gfni vaes vpc
                             lmulqdq rdpid movdiri movdir64b fsr
                             m md_clear serialize ibt flush_l1d
                             arch_capabilities

Virtualization features:
```

Continued **lscpu** output showing virtualization, cache sizes, and CPU vulnerabilities.

```
arch_capabilities
Virtualization features:
  Virtualization:      VT-x
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                 288 KiB (6 instances)
  L1i:                 192 KiB (6 instances)
  L2:                  7.5 MiB (6 instances)
  L3:                  12 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Not affected
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:      Not affected
  Old microcode:        Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed:             Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:    Mitigation; Speculative Store Bypass
                        disabled via prctl
  Spectre v1:           Mitigation; usercopy/swapgs barriers
                        and __user pointer sanitization
  Spectre v2:           Mitigation; Enhanced / Automatic IBRS;
                        IBPB conditional; PBRSE-eIBRS
                        SW sequence; BHI BHI_DIS_S
  Srbds:                Not affected
  Tsa:                  Not affected
  Tsx async abort:      Not affected
  Vmscape:              Not affected
manu@Manasvi:~/os_lab$ |
```

Used the **lsblk** command to list the storage devices (disks, swap, and mount points).

```
manu@Manasvi:~/os_lab$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
manu@Manasvi:~/os_lab$ |
```

Used the **ps** command to display the processes running in the current terminal session.

```
manu@Manasvi:~/os_lab$ ps
  PID TTY          TIME CMD
   338 pts/0    00:00:00 bash
   916 pts/0    00:00:00 ps
manu@Manasvi:~/os_lab$ |
```

Used the **top** command to see a live view of CPU usage, memory, and all running processes.

```
manu@Manasvi:~/os_lab$ top
top - 17:22:41 up 26 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running, 23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7776.6 total, 7172.2 free,  544.0 used,  211.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,   0.0 used,  7232.6 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24164   15420 11620 S   0.0   0.2   0:00.71 systemd
    2 root        20   0    3180    2212  2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    9 root        20   0    3180    2044   1940 S   0.0   0.0   0:00.00 init
   46 root       19  -1   50428   16780 15500 S   0.0   0.2   0:00.18 systemd-journal
   81 systemd+    20   0   22416   14428 11928 S   0.0   0.2   0:00.05 systemd-resolve
   87 root        20   0   35380   12376  9272 S   0.0   0.2   0:00.27 systemd-udevd
  126 root        20   0    2888    1944   1816 S   0.0   0.0   0:00.04 chronyd-starter
  132 root        20   0    4472    3128   2880 S   0.0   0.0   0:00.01 cron
  135 message+   20   0    8816    5360   4676 S   0.0   0.1   0:00.05 dbus-daemon
  154 root        20   0   44096   29764 14620 S   0.0   0.4   0:00.32 networkd-dispat
  159 root        20   0   18436    9360   8116 S   0.0   0.1   0:00.09 systemd-logind
  176 syslog     20   0  220548    5848  4576 S   0.0   0.1   0:00.10 rsyslogd
  227 root        20   0    5492    2720   2532 S   0.0   0.0   0:00.00 agetty
  255 _chrony     20   0   23812   11264   7144 S   0.0   0.1   0:00.11 chronyd
  260 root        20   0  123460   32636 18024 S   0.0   0.4   0:00.24 unattended-upgr
  270 _chrony     20   0   12096    2404   1760 S   0.0   0.0   0:00.02 chronyd
  336 root        20   0    3196    1120    980 S   0.0   0.0   0:00.00 SessionLeader
  337 root        20   0    3196    1252   1108 S   0.0   0.0   0:00.16 Relay(338)
  338 manu        20   0    6268    5524   3856 S   0.0   0.1   0:00.17 bash
  339 root        20   0    8644    5592   4752 S   0.0   0.1   0:00.01 login
  382 manu        20   0   22312   13616 10988 S   0.0   0.2   0:00.10 systemd
  384 manu        20   0   22912    4084   2084 S   0.0   0.1   0:00.00 (sd-pam)
  414 manu        20   0    6136    5432   3864 S   0.0   0.1   0:00.02 bash
  928 manu        20   0    8172    6024   3924 R   0.0   0.1   0:00.00 top
```